9 - K-Nearest Neighbors

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Explain instance-based supervised classification.
- Implement KNN from scratch using a distance metric.
- Evaluate how the choice of k and feature scaling affect predictions.

Background

KNN stores training examples and predicts from nearby examples at inference time. It is simple and interpretable, but prediction cost grows with dataset size and distance is meaningful only when features are sensibly scaled.

Core Concepts

Concept	Meaning in this lab
Feature vector	Numeric representation of an example.
Distance metric	Measures similarity; Euclidean distance is common for continuous features.
k	Number of neighbors voting on the prediction.
Feature scaling	Prevents large-scale features from dominating distance.
Validation	Used to choose hyperparameters without evaluating on the test set.

Guided Procedure

1. Load a labeled dataset such as Iris.

2. Split features and labels into training and test sets.

3. Standardize numeric features using statistics computed on the training set only.

4. Implement Euclidean distance and majority voting.

5. Evaluate several odd values of k and report accuracy.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
import numpy as np
from collections import Counter

def predict_knn(X_train, y_train, x, k=5):
    distances = np.sqrt(((X_train - x) ** 2).sum(axis=1))
    idx = np.argsort(distances)[:k]
    votes = y_train[idx]
    return Counter(votes).most_common(1)[0][0]
```

Lab Exercises

Task 1. Implement KNN without scikit-learn prediction functions.

Task 2. Compare k=1, 3, 5, 7, and 11.

Task 3. Repeat with and without feature standardization.

Task 4. Create a confusion matrix for the best chosen k.

Suggested Deliverables

• Notebook or Python file containing the completed implementation.

• Short result table or output evidence for each required comparison.

• A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. Why can k=1 overfit?

2. Why does scaling matter for distance-based methods?

3. What is the difference between a parameter and a hyperparameter?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.